Mall Customer Segmentation using Machine Learning

Introduction

Mall Customer Segmentation divides customers into groups based on purchasing behavior and income patterns to support business decisions.

Objective

Analyze customer behavior, group similar customers, improve marketing strategies, and identify high-value customers.

Problem Statement

Large customer datasets are difficult to analyze manually. Segmentation helps categorize customers using data-driven methods.

Dataset Attributes

CustomerID, Gender, Age, AnnualIncome_k$, SpendingScore_1_100.

Technologies Used

Python, Pandas, NumPy, Matplotlib, Scikit-learn.

Methodology

Data Collection -> Data Preprocessing -> K-Means Clustering -> Training -> Visualization.

Algorithm

K-Means clustering groups customers into clusters based on similarities.

Advantages

Better customer understanding, personalized marketing, improved retention.

Applications

Retail, shopping malls, e-commerce, banking.

Conclusion

Customer segmentation helps businesses improve decision-making and customer targeting.